

APPTRAIT SOLUTIONS

Practical Assessment: Vibe Coder – Intern / Fresher

Task: Build a Mini HRMS (Human Resource Management System) for a small SaaS company. The objective is not to build a complete enterprise HRMS. We want to evaluate how you take a business requirement, plan the solution, use AI effectively, handle business logic and edge cases, validate AI-generated code, and deliver a working product.

1. User Roles

Implement three types of users:

- **HR/Admin:** Can manage employees and HR operations.
- **Manager:** Can view and manage employees assigned to their team and approve or reject leave requests.
- **Employee:** Can manage or view their own information, attendance and leave requests.

2. Authentication and Authorization

- Login and logout.
- Protected routes.
- Role-based access.

The three roles should have different permissions.

Feature	Admin/HR	Manager	Employee
View all employees	Yes	No	No
Add employee	Yes	No	No
Edit employee	Yes	No	Own profile
View team employees	Yes	Yes	No
Mark attendance	Optional	Optional	Yes
View attendance	All	Team	Own
Apply leave	Yes	Yes	Yes
Approve leave	Yes	Team	No
Reject leave	Yes	Team	No

Important: Permissions should not only be handled by hiding buttons in the frontend. The backend/database layer should also prevent unauthorized actions.

3. Employee Management

HR/Admin should be able to:

- Add employee
- Edit employee
- View employee details
- Search employees
- Filter employees

- Activate/deactivate an employee

Minimum employee information:

Employee ID
Full Name
Email
Phone
Department
Designation
Manager
Joining Date
Employment Status

4. Attendance

- Employees can check in, check out, and view their attendance history.
- HR/Admin can view attendance for all employees and filter by date or employee.
- Managers can view attendance of their own team.

Basic attendance statuses: Present, Absent, Half Day, Leave.

Do not implement biometric integration, GPS/geofencing or complex shift management. Focus on correct application logic.

5. Leave Management

Employees should be able to select leave type, select start and end dates, enter a reason, submit a request, and view previous requests.

Leave statuses: Pending, Approved, Rejected.

Manager

- View leave requests from their team
- Approve leave
- Reject leave
- Add rejection reason

HR/Admin

- View all leave requests
- Approve/reject requests
- Filter by employee, status and date

6. Dashboard

HR/Admin Dashboard

- Total Employees
- Active Employees
- Present Today
- Employees on Leave
- Pending Leave Requests

Manager Dashboard

- Total Team Members
- Present Today

- Team Members on Leave
- Pending Approvals

Employee Dashboard

- Today's attendance status
- Total leave requests
- Pending requests
- Approved/rejected requests
- Recent attendance

Dashboard data must come from actual application data/database and must not be hardcoded.

7. Business Logic and Edge Cases

This is one of the most important parts of the assessment. Before implementing, identify possible problems or ambiguities in the requirements.

- What happens if an employee applies for overlapping leave?
- What happens if the end date is before the start date?
- What happens if an employee applies for leave on a date they already marked attendance?
- Can an employee cancel an already approved or rejected request?
- Can a manager approve a leave request belonging to another team?
- What happens if an employee clicks Check-in twice?
- What happens if an employee tries to Check-out without checking in?
- What happens if an employee tries to check in after being marked on leave?
- Can an employee modify another employee's attendance?

Identify at least **5 edge cases** and implement reasonable handling for them.

8. AI-Assisted Development

AI tools are allowed and encouraged. You may use ChatGPT, Claude, Gemini, Cursor, GitHub Copilot, Kiro, Antigravity, or other AI development tools.

Do not simply ask an AI tool to build the entire application with one prompt and submit the generated result. We want to understand how you work with AI.

AI Development Report

Include a section in your README called **AI Development Process**. Document at least 5 important prompts used during development. For each prompt, explain:

- The prompt
- Why you used it
- The AI's approach/output
- What you accepted
- What you changed or rejected

9. AI Code Review Challenge

Identify at least **2 cases** where AI-generated code needed modification.

For each case, explain:

- What AI generated
- What was wrong
- How you identified the issue
- How you fixed it

10. Planning Before Coding

Before starting development, prepare a short document covering:

- **Requirement Breakdown:** What modules did you identify?
- **Application Flow:** Explain the main user flow from login to core operations.
- **Database Design:** Explain the main entities/tables/collections.
- **Technology Selection:** Explain the frontend, backend, database, authentication and hosting choices, including why you selected them.

11. Technology

You are free to select the technology stack.

- **Frontend:** React.js / Next.js or another suitable framework
- **Backend:** Node.js / Express, Django / DRF, or another suitable backend
- **Database:** PostgreSQL, MongoDB, Supabase, Firebase, or another suitable database

You may use another suitable stack if you can justify your decision. We are more interested in your technical decisions and implementation than a specific framework.

12. Optional Features

Optional features should not be implemented at the cost of core functionality.

- Attendance calendar
- Leave balance
- CSV export
- Employee profile photo
- Pagination
- Advanced search/filter
- Email notifications
- Audit log
- Responsive mobile UI
- Dark mode
- Automated tests

13. UI/UX Expectations

- Clean and understandable interface
- Responsive layout
- Proper navigation
- Form validation
- Loading states
- Empty states

- Error handling
- Success/error notifications

We are not primarily evaluating graphic design. A simple, clean and functional interface is sufficient.

14. Important Security Requirements

- Passwords must not be stored as plain text if you implement your own authentication.
- Protected APIs must not be accessible by unauthorized roles.
- Employee A must not be able to access Employee B's private information by changing an ID in the URL or API request.
- Manager A must not be able to approve leave for another manager's team.
- An employee must not be able to approve their own leave request.

15. Submission Requirements

- **GitHub Repository:** Complete source code.
- **Live Demo:** Deploy the application and provide the URL.
- **README.md:** Project overview, features, tech stack, architecture, database structure, setup instructions, environment variables, demo credentials and known limitations.
- **AI Development Process:** AI prompts and explanations described above.
- **Planning Document:** Initial requirement analysis, application flow, database design, technical decisions and assumptions.

Demo Credentials

HR/Admin
 Email: _____
 Password: _____

Manager
 Email: _____
 Password: _____

Employee
 Email: _____
 Password: _____

We should be able to test the different permissions and workflows using these accounts.

16. Time Limit

Time limit: 2 days from receiving this practical task.

You do not need to build a production-level HRMS. We would rather see 5 properly implemented features than 15 incomplete features.

17. Evaluation Criteria

Area	Weight
Requirement Understanding and Planning	15
Application Architecture	10
Core Functionality	20
Business Logic and Edge Cases	15

Area	Weight
Authentication and Authorization	10
AI Usage and Prompting	10
AI Code Review / Validation	10
UI/UX and Usability	5
Code Quality and Documentation	5
Total	100

18. What We Are Evaluating

We are not testing how much code you can write manually.

As this is a **Vibe Coder** role, AI tools are expected to be part of your development workflow.

We want to see how effectively you can:

Understand → Plan → Prompt → Build → Review → Debug → Test → Improve → Ship

Please be transparent about your AI usage. Using AI is not a disadvantage in this assessment; blindly accepting AI-generated code is.

AppTrait Solutions

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